

ReAct Paper Reproduction Roadmap

Version: Sprint 6 completion checkpoint

Project Objective

Reproduce the prompting protocols from ReAct (ICLR 2023) with inspectable Python control flow, reproducible artifacts, and modern Hugging Face inference. Method fidelity takes priority over batching or model-specific optimization.

Current Scope

Area	Status	Notes
HotpotQA	Implemented and benchmarked	Seven methods, 500 examples each
FEVER	Implemented	Seven methods; full 500-example results TBD
ALFWorld	Not started	Outside Sprint 6
WebShop	Not started	Outside Sprint 6
UI	Not started	Outside Sprint 6

Sprint Status

Sprint	Scope	Status
0	Repository, typed config, logging, CLI	COMPLETE
1	HotpotQA data, metrics, schemas, artifacts	COMPLETE
2	Hugging Face, Standard, CoT	COMPLETE
3	Wikipedia Search/Lookup/Finish, Act	COMPLETE
4	ReAct loop and trajectories	COMPLETE
5	HotpotQA seven-method comparison	FUNCTIONALLY COMPLETED / SUPERSEDED BY EXECUTED BENCHMARK
6	Paper-faithful FEVER reproduction	COMPLETE
7	Research report and failure analysis	NOT STARTED
8	UI	NOT STARTED
9	Optional fine-tuning/extensions	NOT STARTED

Sprint 5 caveat: historical HotpotQA runs used different code versions and batch policies. A controlled rerun is future cleanup. Existing termination, hybrid-path, CoT-SC confidence, and overlap analyses are historical inputs; Sprint 7 itself has not started.

Shared Architecture

```
Benchmark CLI
-> HotpotQA or FEVER dataset adapter
-> shared Standard / CoT / CoT-SC / Act / ReAct / hybrid agents
-> shared Hugging Face provider and batch scheduler
-> independent WikipediaEnvironment per interactive sample
-> task evaluator
-> shared JSON/JSONL/log artifacts
```

FEVER-specific code is limited to its dataset adapter, Appendix C.2 prompt definitions, label parser/normalizer, Accuracy evaluator, and five-step default. No FEVER-specific executable, LLM provider, agent loop, batch runner, Wikipedia tool, or serializer exists.

FEVER Paper-Faithful Setup

- Claim-only input; no FEVER gold evidence or supporting page enters prompts.
- Dataset is the pinned official ReAct data/paper_dev.jsonl file.
- Canonical outputs: SUPPORTS, REFUTES, NOT ENOUGH INFO.
- Exact three manually composed task examples from Appendix C.2.
- Standard, CoT, Act, and ReAct use controlled ablations of those examples.
- Wikipedia action space remains Search, Lookup, and Finish.
- Search returns the first five article sentences or top-five suggestions.
- Repeated Lookup advances through sentence occurrences on the current page.
- FEVER does not terminate early on repeated Search; the five-step budget rules.
- FEVER ReAct default budget is five steps; HotpotQA remains seven.
- CoT-SC samples 21 trajectories at temperature 0.7.
- CoT-SC -> ReAct fallback: winning count below $n/2$.
- ReAct -> CoT-SC fallback: ReAct fails to answer within the step budget.
- Primary FEVER metric is claim-label Accuracy.
- No SFT, LoRA, QLoRA, RL, RAG, embeddings, or alternate retriever.

Appendix C.2 Exemplars

Claim	Gold label
Nikolaj Coster-Waldau worked with the Fox Broadcasting Company.	SUPPORTS
Stranger Things is set in Bloomington, Indiana.	REFUTES
Beautiful reached number two on the Billboard Hot 100 in 2003.	NOT ENOUGH INFO

Prompt version is frozen as fever-appendix-c2-v1. The deterministic tie-break for equal CoT-SC vote counts is the first observed valid normalized label. This is a reproduction implementation detail; the paper does not specify tie handling. For historical compatibility, HotpotQA retains its earlier repeated-Search loop guard; FEVER uses the paper-compatible mode described above.

Paper vs Reproduction

Paper	Reproduction
PaLM-540B	Qwen/Qwen2.5-7B-Instruct
FEVER claim-only	Same
Three FEVER exemplars	Same
Search/Lookup/Finish	Same conceptual action space
CoT-SC n=21	Same
CoT-SC temperature=0.7	Same
FEVER ReAct max steps=5	Same
Primary metric Accuracy	Same
Fine-tuning	Not part of Sprint 6
Batching	Engineering optimization with isolated state

Paper Table 1 Reference

These are PaLM-540B paper results, not project results and not test thresholds.

Method	FEVER Accuracy
Standard	57.1
CoT	56.3
CoT-SC	60.4
Act	58.9
ReAct	60.9
CoT-SC -> ReAct	64.6
ReAct -> CoT-SC	62.0

FEVER Outputs and Metrics

Each run writes `config.json`, `metrics.json`, `predictions.jsonl`, `trajectories.jsonl` for trajectory methods, and `run.log` under `outputs/fever/<method>/<run_id>/`.

Config captures dataset/split, model revision when available, method, sample IDs, seed, batch size, generation parameters, step budget, CoT-SC parameters and threshold, prompt/code versions, and timestamp. Predictions retain canonical labels, invalid parse state, correctness, steps, retrieval tool calls, termination, latency, votes, and hybrid execution path. Metrics report Accuracy, invalid count, class distributions, per-class accuracy, confusion matrix, operational averages, runtime, and termination counts.

Verification Checkpoint

Requirement	Status
Claim-only	PASS
No gold evidence in prompts	PASS
Three Appendix C.2 exemplars	PASS
Standard/CoT/Act/ReAct controlled ablations	PASS
Dense ReAct Thought/Action/Observation	PASS
Search/Lookup/Finish	PASS
CoT-SC n=21 and temperature=0.7	PASS
FEVER ReAct max steps=5	PASS
CoT-SC -> ReAct n/2 heuristic	PASS
ReAct -> CoT-SC failure fallback	PASS
Accuracy primary metric	PASS
Batch state isolation	PASS
HotpotQA regression	PASS
No fine-tuning	PASS

Next Work

Do not automatically start Sprint 7. The next user-driven work is to run the seven 500-example FEVER benchmarks on Colab, then perform a controlled comparison and complete the broader Sprint 7 research report. FEVER scores remain TBD until those runs exist.